

AN **INDIAN** ADAPTATION

Third Edition

Supply Chain Management

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WILEY



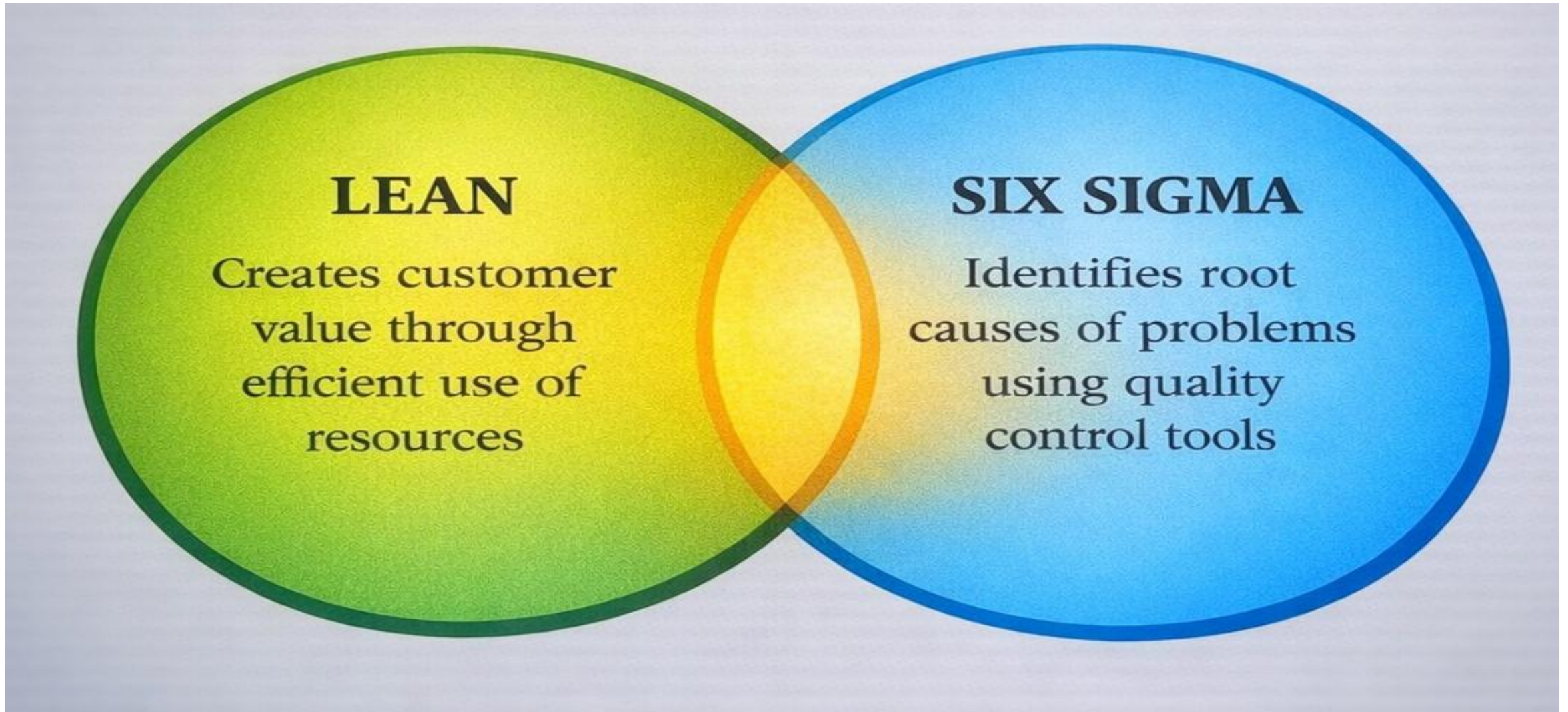
- ***SUBJECT: SUPPLY CHAIN MANAGEMENT STRATEGY***
- ***CHAPTER: 10***
- ***CHAPTER TITLE: Lean Systems and Six-Sigma Quality***
- ***REFERENCE BOOK:***
- ***SUPPLY CHAIN MANAGEMENT (AN INDIAN ADAPTATION), N R SANDERS & S S PADHI, WILEY, 3RD ED. (2023)***

What is Lean?

- ❖ Lean is a management approach for **creating value for the end customer through the most efficient utilization of resources possible.**
- ❖ Originally **pioneered by Toyota in the 1970s**, Lean has been widely adopted in numerous industries.
- ❖ Its implementation often results in **large cost reductions**, **improved quality**, and **increased customer service**.
- ❖ In fact, the benefits are so impressive that Lean has become a standard in many industries, such as auto, aerospace, and the computer industry.
- ❖ One of the most significant is the focus on quality and the use of a **stringent methodology to identify and eliminate root causes of quality problems** called **Six Sigma**.
- ❖ For this reason, the two are often combined through the term Lean Six

LEAN SIX SIGMA :

Combination of two approaches



Six Tenets [fundamental principles] of the Lean Philosophy

1. Elimination of waste

- Eliminate all “non-value”-adding activities.

2. A broad view

- Overall goals should drive all tasks.

3. Simplicity

- The simpler the solution the better.

4. Continuous improvement

- Apply the idea of never-ending improvement.

5. Visibility

- Visible problems are identified and solved.

6. Flexibility

- Quickly adapt to changing customer needs.

Six Tenets [fundamental principles] of the Lean Philosophy

- 1. Elimination of Waste (Removal of Waste):** Actively identifies and removes non-value-adding activities (waste) such as unnecessary transportation, inventory, motion, waiting, and defects.
- 2. A Broad View:** Focuses on optimizing the entire project or system, rather than just individual components or departments, to maximize overall value.
- 3. Simplicity:** Lean focuses on finding simple, practical solutions to problems rather than using costly or complicated methods. The goal is to address the root cause efficiently. Even small changes, like making important information easily visible, can significantly improve workflow and efficiency.

Six Tenets [fundamental principles] of the Lean Philosophy

4. **Continuous Improvement (Kaizen)** : Lean promotes ongoing improvement in all areas, including quality, cost reduction, and waste elimination. The idea is that processes can always be improved. One common method is the **kaizen blitz**, where cross-functional teams work intensively for a few days (2-3 days) to quickly solve specific process problems and achieve low-cost improvements.
5. **Visibility** : Lean emphasizes making problems and waste visible so they can be identified and eliminated. Workplaces should be clean, organized, open, and uncluttered. A visible and orderly environment improves efficiency, reduces confusion, and encourages respect for the workplace.
6. **Flexibility**: Lean systems are designed to adapt quickly to changing customer needs. Flexibility includes adjusting production volumes, producing different product types efficiently, and using multi-skilled workers. Processes are built to remain efficient while accommodating change.

- **Kaizen**: Japanese term that means “change for the better” or “continuous improvement.”
- **Blitz**: German term that means “Lightening fast.”
- **Kaizen Blitz**: A rapid, intensive activity that attacks a specific problem with overwhelming resources to rapidly come to a solution.

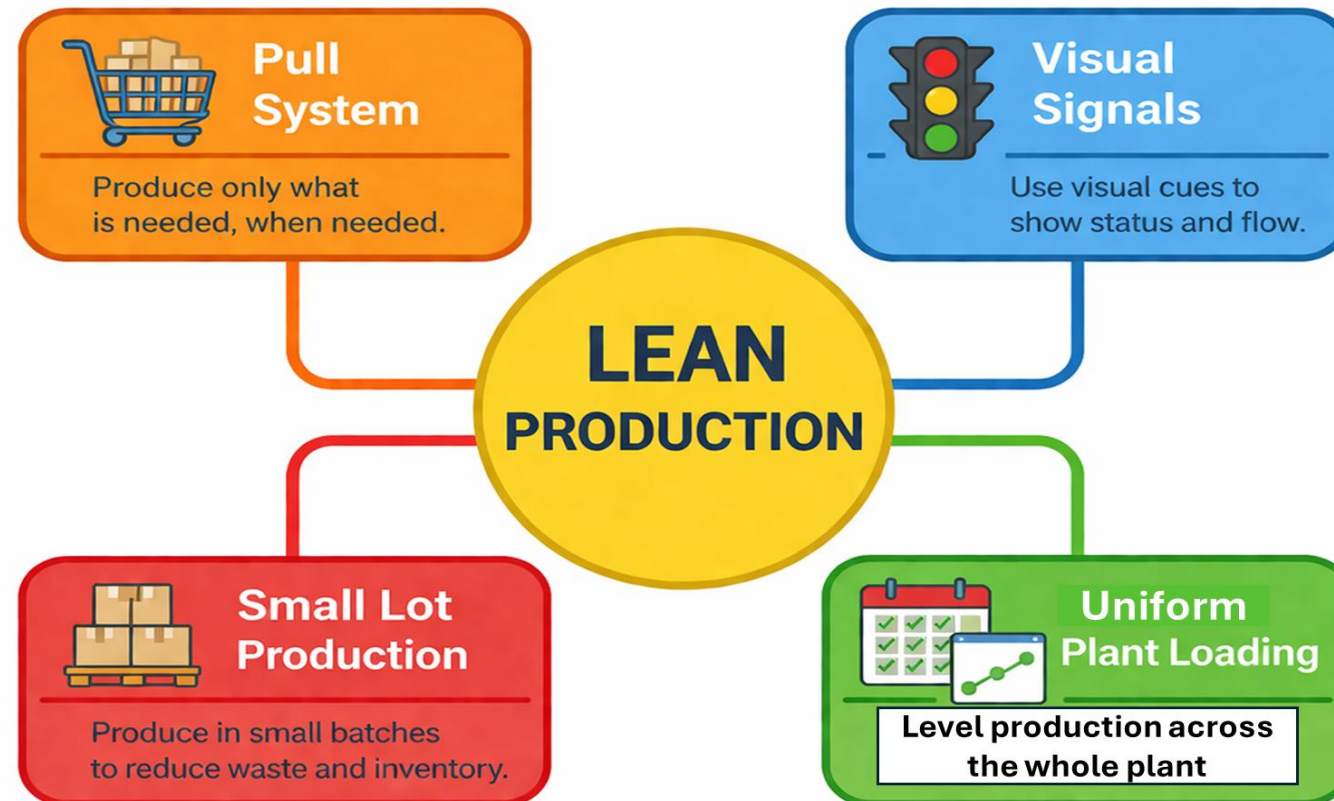
Three Elements of 'Lean'



These **three elements work together** to create a complete Lean system.

1. Lean Production :

- Focuses on the operations and delivery system. Involves developing an efficiently coordinated system that makes it possible to produce the exact products desired and deliver them in the right quantities to where they are needed **just in time (JIT)**.



1. Lean Production :

- Focuses on the operations and delivery system
- Involves developing an efficiently coordinated system that makes it possible to produce the exact products desired and deliver them in the right quantities to where they are needed **just in time (JIT)**.
- Known as a **pull production system** that keeps minimal levels of inventory, relies on efficiencies gained by repetition, elimination of all unnecessary steps and motions, and use of visual signals.

Pull System



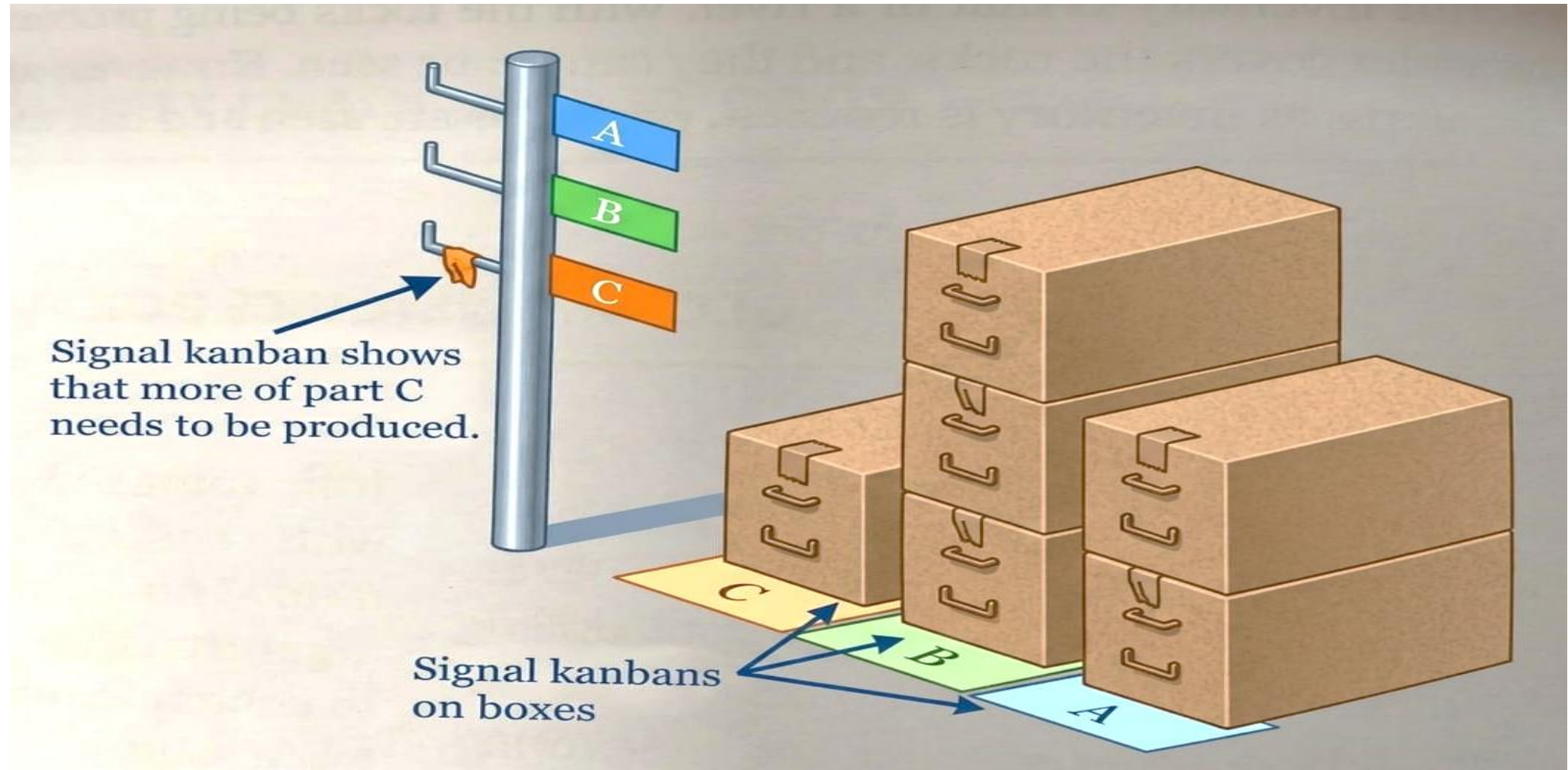
- **Push System** – Production based on forecasting and schedules, "pushing" products to the next station regardless of current demand.
- **Pull System** - produce and move products, i.e., produces only when required in the next step in the supply chain.
- Each stage in the supply chain requests quantities needed from the previous stage in the chain.
- If products are not requested, they are not produced and no excess inventory is generated.
- Lean views inventory as a cost that provides no value and should be eliminated, and thus helps maintain low inventory levels, minimising storage costs and waste.
- Improves flexibility and responsiveness to changes in customer needs.
- Reduces lead time and inefficiencies by synchronizing production and supply chain activities.

Visual signals



- Visual signals are essential for **communication between work centers and supply chain partners** in a pull system.
- A common signaling tool is the **kanban** (means 'signal' in Japanese) , which acts as a card or signal to trigger production.
- Kanban cards contain important details such as **product name, part number, and required quantity**.
- Production is **authorized only** when a kanban signal is present, ensuring no overproduction occurs.
- Visual signals can take different forms such as **kanban cards, squares on the floor (kanban squares), or flags (signal kanban)**.
- In supply chains, **supplier kanbans coordinate deliveries**, allowing suppliers to replenish materials and pick up empty containers efficiently.

Visual signals in lean production using Kanban tool



Small Lot Production



- Small lot production means **producing in small quantities** (e.g., 10 units instead of 1,000) at a time.
- It helps in **reducing inventory and eliminating excess processing**, leading to less waste.
- It **shortens manufacturing lead time**, since smaller batches take less time to produce.
- It requires **reduced setup time**, making it easier to switch between different products quickly.
- It improves **flexibility**, allowing companies to respond quickly to changing customer demands.
- It **enhances product variety and customization**, as different products can be produced in multiple small batches.

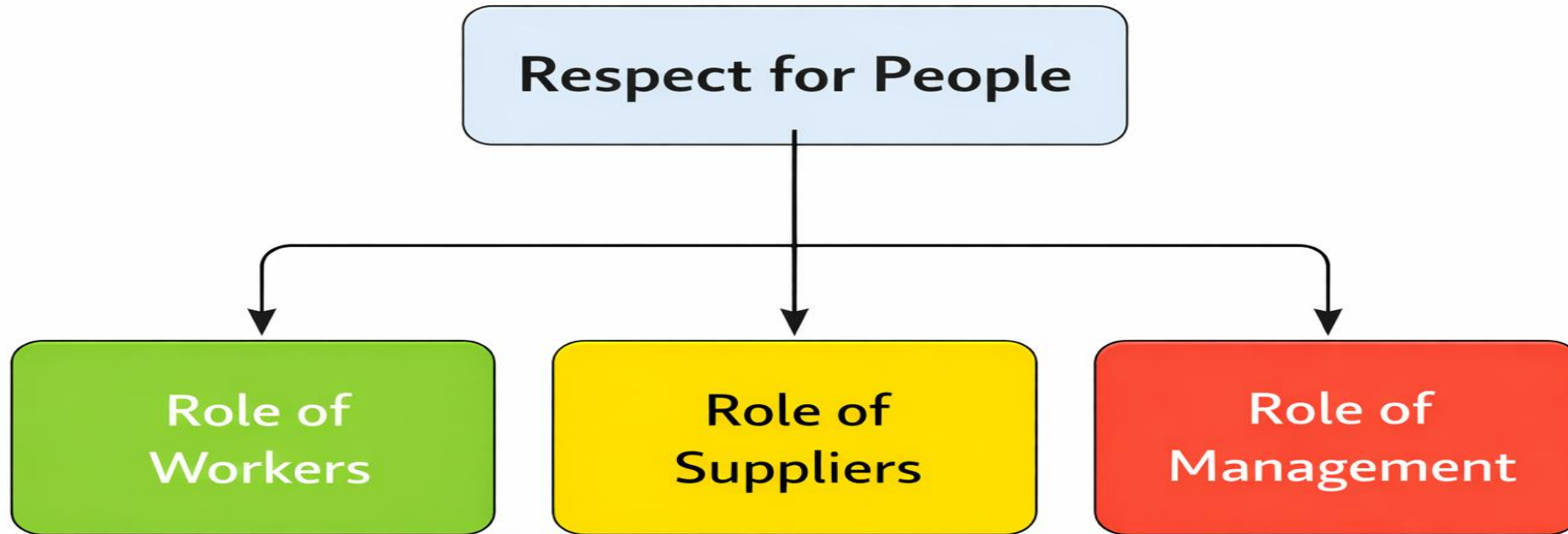
Uniform Plant Loading



- Uniform Plant Loading (also called ‘**Leveling**’) means **maintaining a steady and consistent production schedule over time** by minimizing sudden changes in demand.
- It helps to **reduce disruptions in production** caused by sudden increases or decreases in demand.
- It **prevents inefficiency and waste** that occur due to fluctuating production schedules.
- It involves setting a **stable production plan** over a specific period.
- It ensures a **smooth and balanced workflow**, avoiding overload or idle time
- It leads to **better utilization of resources** like labor and machinery.

Respect for People :

Treating all employees with respect



Role of Workers

- Workers in Lean organizations are **actively engaged in company goals**, not just performing repetitive tasks.
- Lean emphasizes **cross-functional skills**, allowing workers to operate multiple machines and handle diverse tasks.
- Employees are **responsible for improving processes, monitoring quality, and fixing problems**, not just executing work.
- Workers are given **autonomy, and their opinions are valued**, especially since they are closest to production issues.
- Through **jidoka**, workers have the authority to **stop production to address quality problems immediately**, promoting ownership and accountability.
- Workers collaborate in **quality circles (teams of 5–12 people)** to regularly identify, discuss, and solve quality issues and share solutions with management.
 - **Jidoka**: A Japanese term that means automation with human intelligence
 - **Jidoka** originated in the early 1900s when Sakichi Toyoda, founder of the Toyota Group, invented a textile loom that stopped automatically when any thread broke.

Role of Management

- **Management must treat employees as valuable assets**, not just labor, **recognizing** their **skills, knowledge, and contributions**.
- They are responsible for **creating a culture of trust and respect**, where workers feel safe to share ideas and report problems without fear.
- Managers should **empower employees** by giving them authority (e.g., to stop production) and involving them in decision-making.
- They must **support continuous improvement (Kaizen)** by encouraging suggestions and providing time and resources for problem-solving.
- Management provides **training and skill development**, helping workers become multi-skilled and more capable.
- They should **listen actively to employees' ideas and feedback** and act on them, reinforcing that worker input is valued.
- Managers are responsible for **removing obstacles and providing support**, rather than blaming workers for issues.
- They must **promote teamwork**, such as quality circles, and ensure collaboration across the organization.

Role of Suppliers

- **Suppliers are treated as long-term partners**, not just vendors, building mutual trust and cooperation.
- They are **involved early in product design and improvement processes**, contributing their expertise.
- Lean organizations **work closely with suppliers to improve quality and efficiency**, rather than blaming them for defects.
- Suppliers are expected to **deliver high-quality materials consistently**, supporting smooth production flow.
- There is an emphasis on **open communication and information sharing** to solve problems collaboratively.
- Companies help suppliers **develop their capabilities** (training, process improvement) to meet Lean standards.
- Suppliers are encouraged to **adopt Lean practices themselves**, ensuring alignment across the supply chain.

Total Quality Management (TQM)

- TQM is integral to Lean as it identifies customer quality standards and serves to eliminate waste, such as defects, scrap, and rework.
- TQM focuses on **continuous improvement of quality** in all processes, products, and services.
- It involves **everyone in the organization**, from top management to workers, in improving quality.
- TQM **emphasises customer satisfaction as the primary goal** by focusing on **preventing defects** rather than detecting them later, by using **data, teamwork, & systematic problem-solving** to improve process.

Statistical Quality Control (SQC)

- SQC is the **use of statistical tools** to measure product and process quality.
- These tools provide quantifiable measures that verify quality standards are being met and can be divided into three broad categories:
 - **Descriptive statistics**: used to describe quality characteristics.
 - **Statistical process control (SPC)**: a statistical process used to verify that the production process is functioning as specified and involves inspecting a random sample of the output from the process, deciding whether the process is producing products with characteristics that fall within the acceptable range.
 - **Acceptance Sampling**: It is the process of randomly sampling a batch of goods, deciding whether to accept the batch based on the results.

Variation and its Sources

Variation: 'Deviation from the standard norms'

- All processes contain natural variation caused by different factors.
- Variation that can be identified and controlled is called **assignable variation**, such as poor worker training, machine misalignment, or low supplier quality which can usually be corrected.
- On the other hand, **common or random variation** is inherent in the process and cannot be completely eliminated.
- Examples include slight differences in the fill level of soft drink bottles or the number of blueberries in muffins.

Since no two products are exactly identical, quality control focuses on measuring and monitoring random variation to ensure it remains within acceptable limits.

Process

- The ‘**process**’ is a **set of interrelated activities** that convert the inputs such as equipment, materials, people, methods, and environment into desirable output.
“**No two things are alike and variation is law of nature and exists in all processes**”
- Variation in a product and process can be always measured.
- The **quality of the process** is measured by **the output of product quality characteristics**.
- If product characteristics/specifications conform to engineering/customer requirements, then the **process is considered stable/capable** and performing well as per set standards.
- A process is considered **stable if there are only Common Causes of Variation** (chance or natural) present in the process.

Process Capability

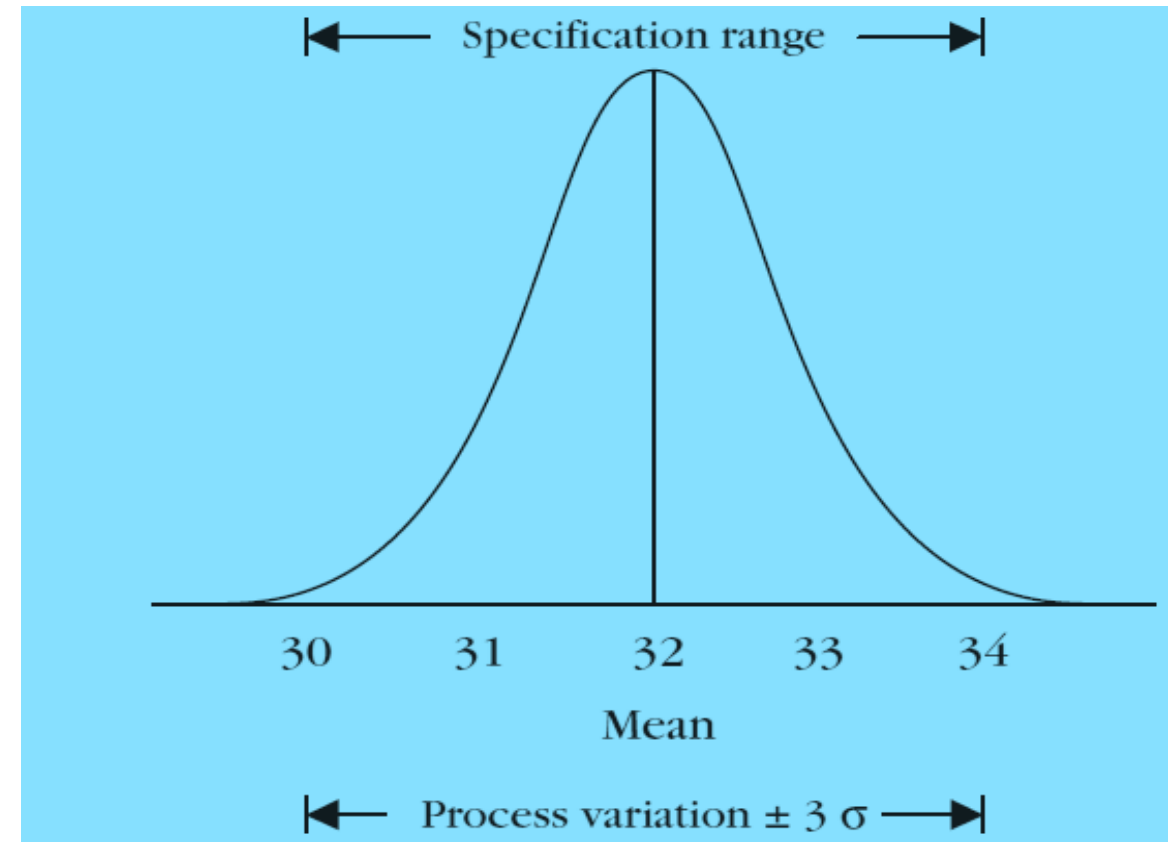
- **Process capability** is the evaluation of a process as to its ability to meet certain quality standards.
- **A quality program first checks whether a process is capable of meeting desired product specifications or tolerances.**
- **Product specifications** define the acceptable range of quality characteristics, such as dimensions, weight, or service standards, based on design requirements and customer expectations.
 - For example, a part specified as 10 inches $\pm$ 0.5 must fall between 9.5 and 10.5 inches to be acceptable.
- Since every process has natural variation, a process is considered capable only if its variation stays within the specified limits.

Process capability means comparing process variation with product specifications to determine whether a process can consistently produce products within specification limits.

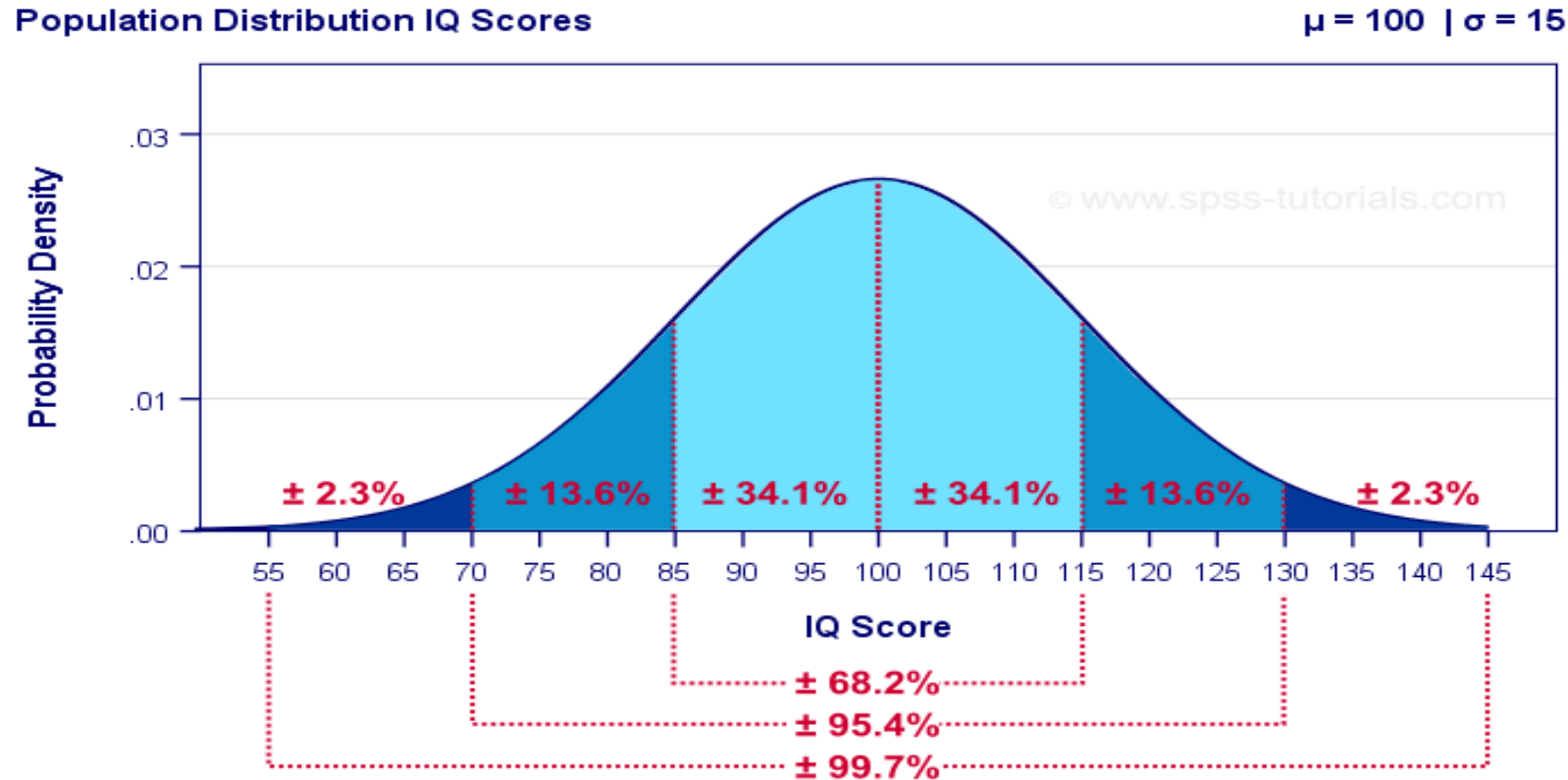
Higher capability values indicate better and more reliable process performance.

Process Capability

- ❖ Consider Pasta Mania, a company that produces pasta sauces in glass jars specified at 32 ounces ± 2 ounces.
- ❖ Therefore, the product specification range is between 30 and 34 ounces.
- ❖ The given figure shows a process (3 standard deviations) where product volumes fall between 30 and 34 ounces.
- ❖ Assuming a normal distribution, 99.74% of products fall in the range of 3 standard deviations.
- ❖ It is observed that process variation matches product specifications.
- ❖ The process is capable of producing products within the set specifications.



Process Capability Index



- $1\sigma \Rightarrow$ Approx. 68.2% cases fall within the specified range ' $\mu + \sigma$ ' and ' $\mu - \sigma$ '
- $2\sigma \Rightarrow$ Approx. 95.4% cases fall within the specified range ' $\mu + 2\sigma$ ' and ' $\mu - 2\sigma$ '
- $3\sigma \Rightarrow$ Approx. 99.7% cases fall within the specified range ' $\mu + 3\sigma$ ' and ' $\mu - 3\sigma$ '

Measuring Process Capability

Process capability is measured using an index, the Process capability Index (C_p).

$$C_p = \frac{\text{Product specification range}}{\text{Process variation range}}$$

$$C_p = \frac{USL - LSL}{6\sigma}$$

USL: Upper Specification Limit

LSL: Lower Specification Limit

Process Capability Index

$$C_p = \frac{USL - LSL}{6\sigma}$$

- ❖ The product specification range is the difference between the upper specification limit (USL) and the lower specification limit (LSL) of the process.
- ❖ The process variation range —the denominator— is computed as 6 standard deviations (6σ) of the process being monitored.
- ❖ In a normal distribution, almost all process output lies within **± 3 standard deviations** from the mean.
- ❖ For a normally distributed process:
 - About **68.27%** of observations lie within $\pm 1\sigma$
 - About **95.45%** lie within $\pm 2\sigma$
 - About **99.73%** lie within $\pm 3\sigma$
- ❖ Thus, nearly all process values fall between: $\mu - 3\sigma$ to $\mu + 3\sigma$
- ❖ The total spread becomes: $(\mu + 3\sigma) - (\mu - 3\sigma) = 6\sigma$
- ❖ 6σ is used because 99.73% of the products fall within 3 standard deviations of the distribution—a total of 6 standard deviations.

Process Capability Index

$$C_p = \frac{USL - LSL}{6\sigma}$$

INTERPRETATION OF THE RESULT

- $C_p = 1$: Process variation just meets specifications, and the process is “barely (minimally) capable.”
- $C_p < 1$: Process variation exceeds the specification range, and the process is not capable of producing products within specification, i.e., defects are inevitable.
- $C_p > 1$: Process variation is less than the specification range, and the process exceeds minimal capability.

Process Capability Index

$$C_p = \frac{USL - LSL}{6\sigma}$$

Pasta Mania is evaluating three different machines for their capability in filling jars with its famous pasta sauce. The product specification range is set between 30 and 34 ounces. Given the standard deviations for each machine provided below, determine which of the machines are capable of producing within specifications.

Machine	A	B	C
Standard Deviation	0.6	0.7	1.2

To determine the capability of each machine, we need to divide the specification width ($UCL - LCL = 34 - 30 = 4$) by 6σ for each machine:

Machine	σ	$USL - LSL$	6σ	$C_p = \frac{USL - LSL}{6\sigma}$
A	0.6	4	3.6	1.11
B	0.7	4	4.2	0.95
C	1.2	4	7.2	0.55

Looking at the C_p values, only **machine A** is capable of filling bottles within specifications because it is the only machine that has a C_p value above 1.